GMI German-Malaysian Institute

DIPLOMA PROGRAMME

INDIVIDUAL CASE STUDY

Academic Period: July 2026

Program	:	**CYBER SECURITY TECHNOLOGY (CBS)**
Course	:	**NETWORK MANAGEMENT & MONITORING SYSTEMS**
Code	:	**NTC 1052**

Title	:	**FIXING EDUTECH'S NETWORK CHAOS**

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CASE STUDY ASSIGNMENT: Fixing EduTech's Network Chaos.

Background Scenario

You've just been hired as part of the Network Engineering team at **EduTech Institute**.

Over the past few years, the school expanded from one main campus building to three separate campuses. Sounds great, right? The problem is that the IT department's documentation didn't grow with it. Right now, the school's entire network inventory around **1,500 devices,** including routers, switches, access points, and file servers is tracked on a single, outdated Excel sheet that nobody has touched in two years.

Things have started falling apart:

- **The 3-Hour Search:** Last week, a core switch failed on Campus B. The team spent three frustrating hours running around trying to physically find the switch and figure out what software version it was running before they could even attempt a fix.
- **The Unwanted Guest:** An attacker recently breached the network through an old bridge device running **Software Version 'A'**. Because it wasn't recorded on the old spreadsheet, nobody knew it existed or that it needed a security patch.
- **Slow Connections:** Students and staff keep complaining about lag. Different departments have been setting up their own file servers without telling IT, leading to IP address conflicts and redundant services.

Management has finally given your team the green light to implement a proper **Network Management System (NMS)** and clear up this mess.

Your Task

As a team, you need to put together a solid plan for EduTech's new Configuration Management setup. You'll submit a brief written report and present your pitch to me (acting as EduTech's Chief Technology Officer).

Part 1: How will you map the network?

1. **Choosing a Discovery Method:** You need to find all 1,500 devices on the network. Will you use a **Query-Based (Active Ping)** scan or a **Protocol-Based** approach? Explain why you made your choice, and be honest about **one upside** and **one downside** of that approach.
2. **Visual Mapping:** Explain how using **Automapping** will save your team from wasting hours during the next hardware outage on Campus B.

Part 2: Upgrading how we store data

3. **Ditching the Spreadsheet:** Give **two solid reasons** why sticking with flat text files or spreadsheets is a recipe for disaster in a dynamic 1,500-device network.
4. **Moving to a Database (DBMS):** How will moving our inventory into a DBMS make life easier for the engineering team? (Mention **two advantages**).

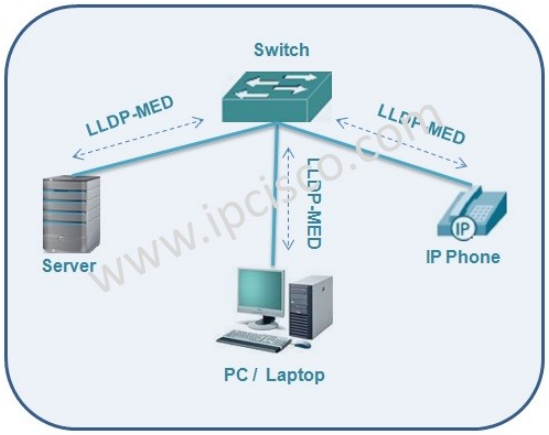
Part 3: Fixing security and daily glitches

5. **The Security Audit:** Write the exact **SQL query** you'll run on the new system to instantly pull up every vulnerable **bridge device running Software Version 'A'** so we can patch them immediately.
6. **Remote Fixes:** A switch port on Campus C is throwing constant errors and dragging down performance. Walk me through how your team will troubleshoot and disable or fix that interface **from your desk**, without having to drive over to Campus C.

Part 1: How will you map the network?

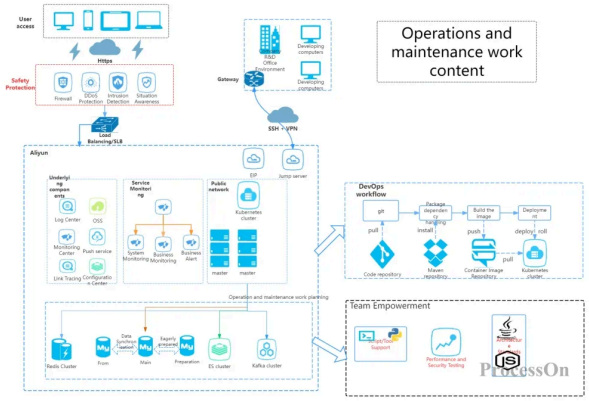
Discovery Method : Protocol-Based Approach

Instead of blindly pinging IP address ranges, protocol based discovery queries device neighborhood tables and system parameters. This allows the NMS to understand not just that a device exists but how devices are interconnected at Layer 2 and Layer 3.



The upside is the topology intelligence, It will basically map physical port to port relationships and device dependencies, providing actual structural layout rather than just a list of active IP addresses.

But the downside is it requires pre configured SNMP credentials/community strings on all managed switches or routers. Unmanaged switches or rogue devices lacking these protocols won't respond to protocol queries without an auxiliary ping sweep.



How it works is basically when a switch on Campus B drops, Automapping instantly highlights the failed device in red on the graphical map.

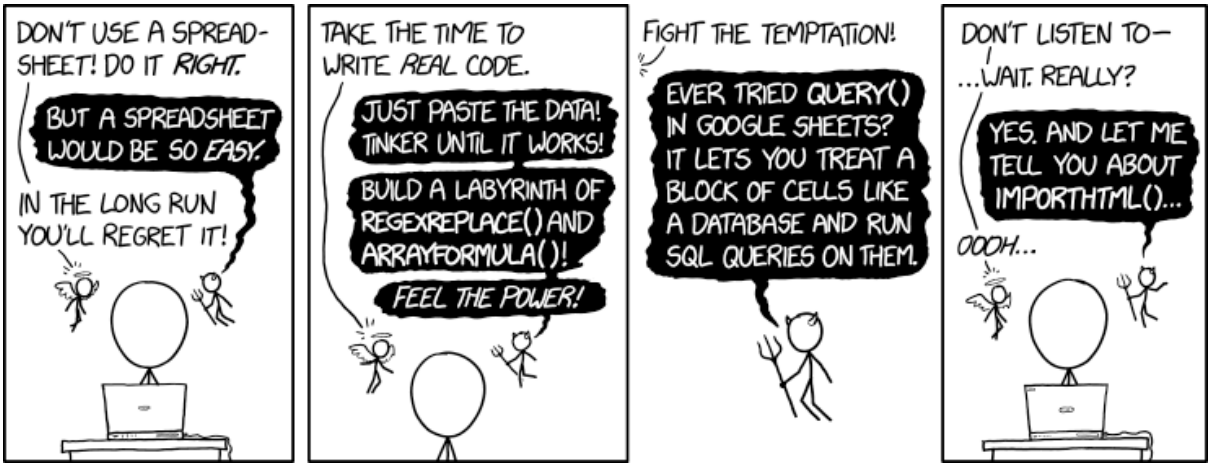
Automapping attaches device inventory metadata directly to the visual note, telling technicians exactly where to walk. For example Campus B -> Building 2 -> MDF Closet -> Rack 03 -> Unit 12.

Part 2: Upgrading how we store data

Reason for Phasing Out Spreadsheets

The reason why Spreadsheet is hard to work with, basically Spreadsheet is lacking in concurrency or audit control. Maintaining network data on spreadsheets in a multi-user engineering environment leads to version sprawl, file locking issues and unrecorded manual edits, while allowing shadow IT deployments such as rogue file servers to bypass tracking entirely.

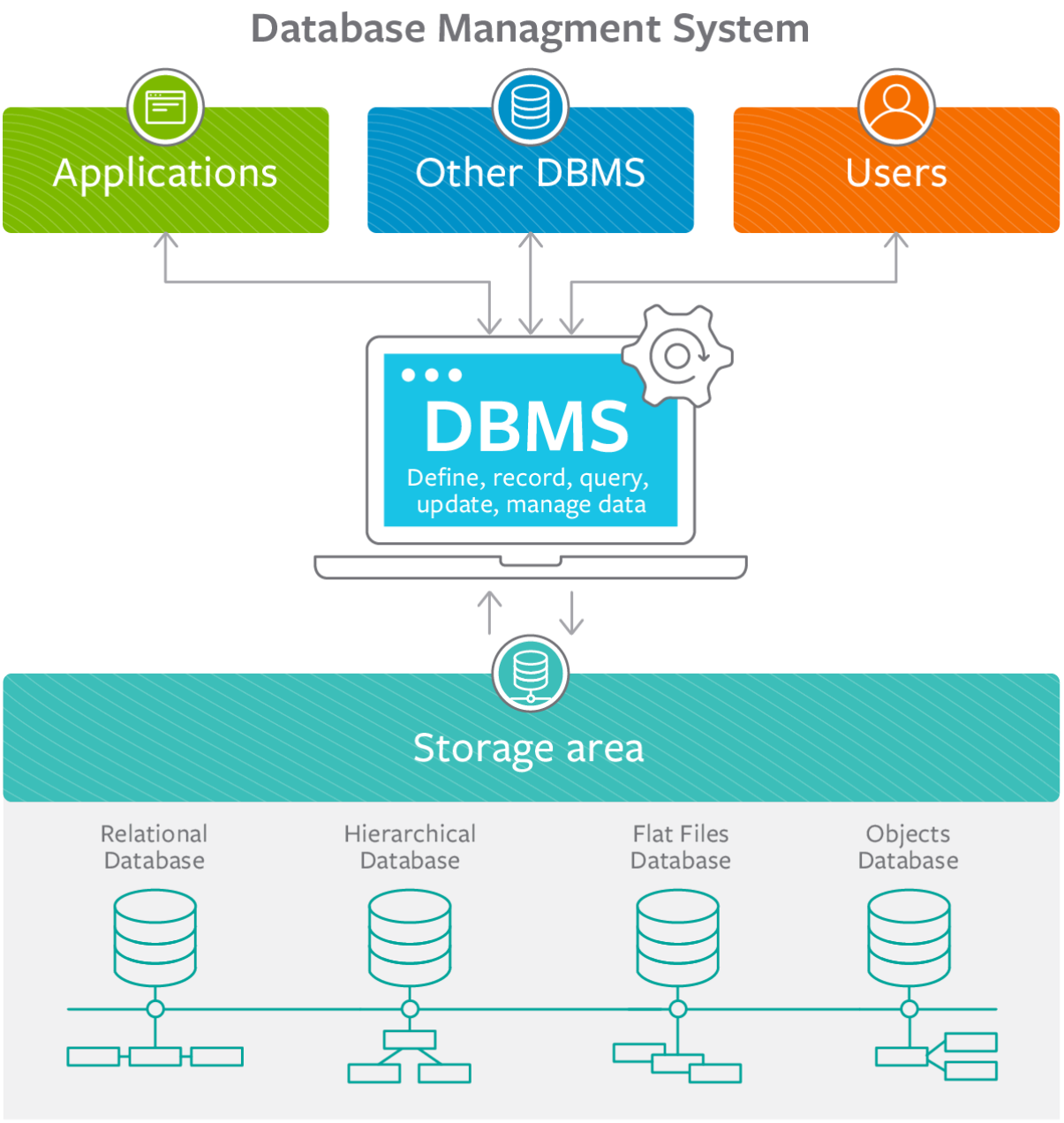
Other than that, Spreadsheets are inherently static documents that cannot automatically poll devices, trigger real-time alerts or update inventory status which leaves vulnerable hardware running unpatched software completely hidden until a system failure or security breach occurs. Basically Spreadsheets = Stress-sheets.



Advantages of Moving to a Database Management System (DBMS)

The main advantage of upgrading to a DBMS is its seamless automated NMS integration. Basically, a DBMS dynamically syncs with active scanning agents to automatically populate firmware updates, uptime statuses, and IP leases in real-time without requiring any error-prone manual data entry.

Other than that, a DBMS provides robust programmatic querying and strict governance. It supports fast filtering, multi-user role-based access control (RBAC), and transactional integrity, which ensures complete data consistency and security across all 1,500+ network endpoints. Basically DBMS = Stress-Free Control.



bmc

Part 3: Fixing security and daily glitches

Fixing Security and Daily Glitches

To query the database for all vulnerable bridge devices running Software Version A :

Reference from (https://www.programiz.com/sql/select)

```
SELECT
  device_id,
  device_name,
  ip_address,
  mac_address,
  location,
  software_version
FROM
  network_inventory
WHERE
  device_type = 'Bridge'
  AND software_version = 'Version A';
```

Campus C Port Remediation

1) Identify NMS Dashboard Alert
2) Diagnose SNMP Diagnostics
3) Isolate Remote Shutdown
4) Reconfigure

- Identify & Access by locating the switch IP on Campus C via the NMS dashboard. Then open a secure remote session (ssh) into the switch management IP from your desk.
- Diagnose interface errors by running interface diagnostics commands to inspect the error counters:

  ```
  show interfaces status
  show interfaces GigabitEthernet1/0/12 counters errors
  ```

  And look for high counts of CRC errors, frame errors or late collisions indicating a duplex mismatch or failing cable.

- Isolate and temporarily shutdown the noisy interface from CLI to prevent error propagation across the network :

  ```
  configure terminal
  Interface G1/0/12
  shutdown
  ```

- Reconfigure it if the speed or duplex negotiation issue, manually force parameters :

  ```
  speed 1000
  duplex full
  no shutdown
  ```

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